

FusionGuardNet: Audio Deepfake Detection

Using WavLM, Whisper, and Nes2Net

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Project Report

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Abstract

The rapid advancement of generative AI has made synthetic speech increasingly indistinguishable from natural human voice, posing serious threats to voice authentication systems and audio-based communication. This project presents **FusionGuardNet**, an audio deepfake detection system that combines complementary representations from two large pre-trained models—**WavLM** and **Whisper**—processed by a **Nes2Net** classifier.

WavLM captures fine-grained acoustic and speaker-level properties of the audio signal, while Whisper encodes phonetic and prosodic structure derived from its speech recognition training. Both encoders are kept fully frozen, and their 768-dimensional frame-level embeddings are combined through a lightweight learnable weighted fusion layer before being passed to the classifier. This offline extraction strategy eliminates redundant encoder inference during training and keeps the system practical on a single GPU.

FusionGuardNet is evaluated on two benchmark configurations. On **Dataset 1** (ASVspoof Combined, 10,536 test samples), the model achieves **99.18% accuracy** with a perfectly symmetric error profile of 43 false positives and 43 false negatives. On **Dataset 2** (ASVspoof-FoR Extended, 17,467 test samples), it reaches **99.34% accuracy** and an **Equal Error Rate of 0.60%**, demonstrating that performance remains stable under a broader and more diverse set of spoofing conditions.

The results support the central hypothesis that fusing acoustic and phonetic-linguistic representations yields a more robust detector than either modality alone, while introducing only 1,536 additional trainable parameters over the single-encoder baseline.

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1 Introduction

1.1 Challenges in Audio-Based Deepfake Detection

Recent advances in generative AI have made it significantly easier to produce synthetic speech that sounds indistinguishable from a real human voice [1, 2]. Modern text-to-speech (TTS) and voice conversion (VC) systems can now generate highly convincing audio with minimal input, enabling an attacker to clone a speaker’s voice from only a short recording [3]. As these technologies continue to improve, the boundary between authentic and synthetic speech becomes increasingly difficult to identify using human perception alone.

These audio deepfakes pose a growing security threat [4, 5]. They can be used to bypass voice authentication systems, impersonate individuals in phone calls, manipulate financial transactions through social engineering, or spread disinformation through fabricated recordings [6]. The risk is especially significant because audio is often treated as a trusted communication channel in banking, customer service, journalism, legal evidence, and personal communication.

The research community has responded through the ASVspoof challenge series [7, 8, 9], which has run since 2015 and provides standardized benchmarks and evaluation protocols for anti-spoofing systems. Building an effective detector is not straightforward: classical approaches relied on handcrafted features [10] and tend to miss the subtle artifacts that modern synthesis methods introduce, while generalization to unseen attack types remains a major open challenge [11].

Fake audio can fail in more than one dimension simultaneously. Some artifacts are acoustic—unnatural spectral patterns, phase inconsistencies, or waveform irregularities. Others are phonetic or prosodic—irregular rhythm, weak co-articulation, or unnatural intonation [12, 13]. A detector relying on a single feature type is vulnerable to synthesis systems that do not happen to trigger that specific artifact [14].

Our project addresses this challenge by exploring how combining different types of pre-trained audio representations can produce a more robust and accurate deepfake detector.

1.2 Project Goals: Improving Spoofing Classification Using Pre-trained Models

The main goal of this project is to build a system that reliably distinguishes between real and synthetic speech and validate it on two established public benchmark configurations. Rather than designing handcrafted features or training from scratch, we leverage large pre-trained models [15, 16] that have already learned rich speech representations from massive audio data.

A key design decision is the use of **offline feature extraction**: all audio representations are extracted and saved to disk before training begins. This cleanly separates the costly encoding step from the iterative training loop, dramatically reduces GPU memory pressure, and makes it practical to experiment with different classifier architectures without re-running large encoder inference.

The end result is a complete detection pipeline trained and evaluated across two dataset configurations: **ASVspoof 2019 LA** [7] combined with **ASVspoof5 (2024)** [8] as Dataset 1, and an extended version adding **Fake-or-Real (for-norm)** [22] as Dataset 2.

1.3 Our Contribution: Integrating Whisper with WavLM in the Nes2Net Architecture

Our main contribution is the design and implementation of **FusionGuardNet**, which combines **WavLM** [15] and **Whisper** [16] to analyze the same audio from two complementary perspectives. WavLM captures low-level acoustic signal patterns, while Whisper encodes phonetic and prosodic structure. Both representations are extracted offline, combined through a learnable weighted fusion layer, and passed into a shared **Nes2Net** [33] classifier.

Both encoders remain fully frozen during training. Only the fusion layer and the Nes2Net backbone are updated, keeping the trainable parameter count small and preventing catastrophic forgetting. WavLM and Whisper both produce 768-dimensional frame-level embeddings, so the two sequences can be fused without any additional projection layer.

On Dataset 1, the system achieves **99.18% test accuracy** with a perfectly symmetric error profile (FP = 43, FN = 43). On Dataset 2, it reaches **99.34% accuracy** and an **EER of 0.60%**.

2 Related Work

2.1 From Classical Methods to Pre-trained Representations

Spoofing refers to the deliberate use of synthesized, replayed, or voice-converted speech to deceive an automatic speaker verification system. Early anti-spoofing systems relied on handcrafted features [10] combined with classical classifiers. As neural TTS improved [1, 3], these fixed representations became insufficient and the field moved toward deep learning [2].

The most effective direction has been large self-supervised models pretrained on vast natural speech corpora [1, 18], which generalize better to unseen synthesis methods [11, 25]. Two such models form the backbone of our approach.

WavLM [15] is developed by Microsoft and trained with a masked speech prediction task applied to noise-mixed utterances, forcing it to build robust acoustic representations. It is built on the wav2vec 2.0 framework [26], using a CNN waveform encoder followed by a Transformer stack with Gated Relative Position Bias. Unlike HuBERT [27] or wav2vec 2.0, WavLM’s denoising pre-training objective makes its representations particularly robust to signal-level perturbations. The variant we use, `microsoft/wavlm-base-plus`, produces frame-level embeddings of dimension **768** at 20 ms resolution [15].

Whisper [16] is developed by OpenAI and trained on 680,000 hours of transcribed audio. Because its objective is transcription, it learns phonetic and linguistic structure rather than raw acoustic properties—a perspective directly relevant to deepfake detection [28], since

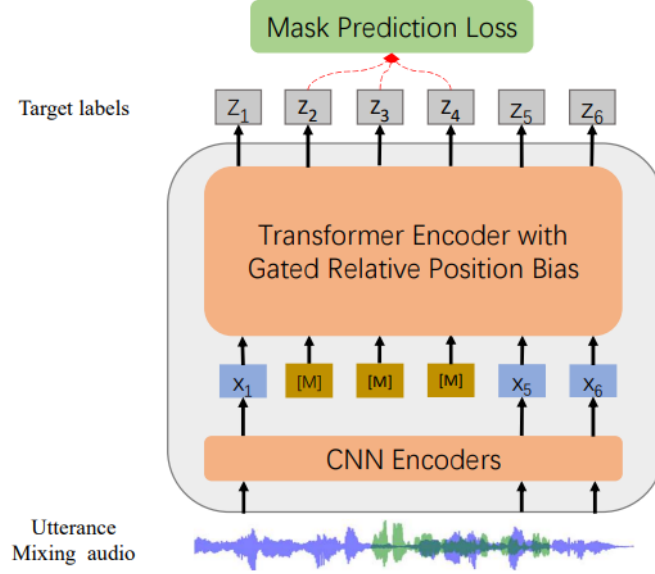


Figure 1: WavLM pre-training architecture. CNN encoders extract local features from raw audio; the Transformer encoder with Gated Relative Position Bias contextualises them. A subset of frames is masked ([M]) and the model is trained to predict target representations Z_i from context via Mask Prediction Loss [15].

synthetic speech often contains subtle phonetic irregularities that acoustic models may overlook. Architecturally, Whisper follows an encoder-decoder design; we discard the decoder entirely and use only the encoder, which also produces **768-dimensional** embeddings [16]. This dimensional alignment with WavLM means no projection layer is needed between the two branches. Together, WavLM and Whisper offer complementary views of the same signal—one acoustic, one phonetic—which is the core idea behind our fusion approach [23, 24].

2.2 Classification Architecture: Nes2Net

Standard ResNet [29] introduced residual connections that allow gradients to flow cleanly through deep networks. Res2Net [30] extended this by splitting feature channels into hierarchical subsets, enabling multi-scale feature capture in a single forward pass—useful for audio because spoofing artifacts appear at different time resolutions [31, 32].

Nes2Net [33] (Liu et al., 2025) nests an additional residual connection inside each Res2Net block, creating a two-level hierarchy of residual pathways. An inner shortcut preserves fine-grained information while the outer one propagates more abstract representations. Each block also incorporates a Squeeze-and-Excitation (SE) module for channel-wise recalibration. A key practical advantage is that Nes2Net accepts the high-dimensional output of speech foundation models directly, without a dimensionality reduction step, followed by global pooling and a linear classification head.

3 Dataset and Preprocessing

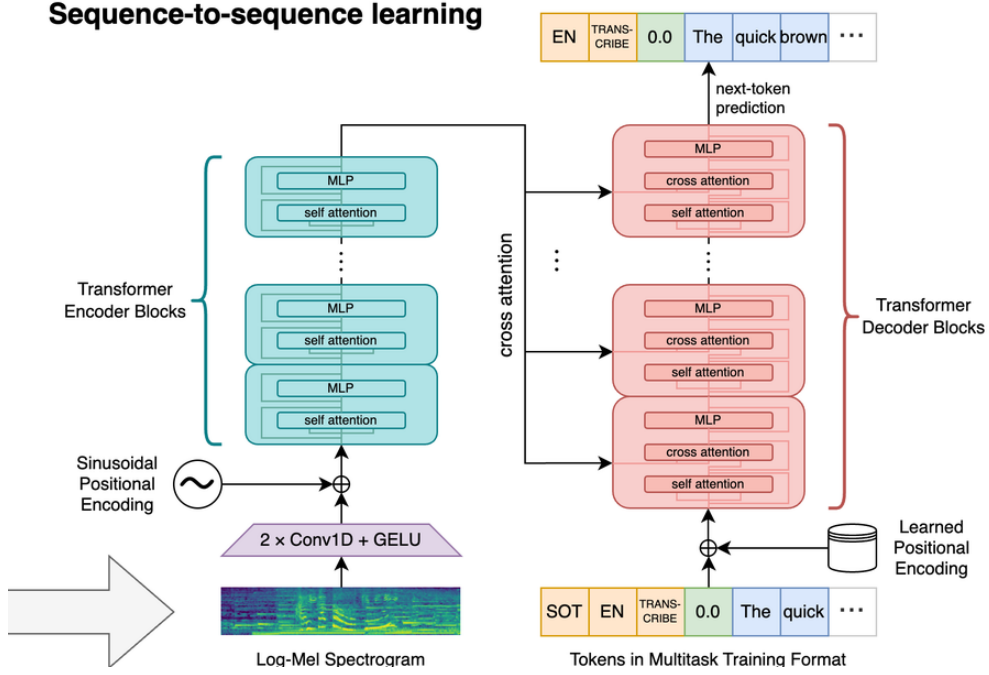


Figure 2: Whisper sequence-to-sequence architecture [16]. The encoder (left, teal blocks) processes a Log-Mel Spectrogram through 2xConv1D + GELU and stacked Transformer blocks. In FusionGuardNet only the encoder is used; the decoder (right, pink blocks) is discarded after feature extraction.

3.1 The Selected Datasets

We conducted experiments on two dataset configurations of increasing scale.

3.1.1 ASVspoof 2019 Logical Access (LA)

ASVspoof 2019 LA [7] is one of the most widely adopted benchmarks in speech anti-spoofing. It contains recordings from 19 different TTS and VC algorithms alongside genuine speech collected in a controlled environment, distributed in FLAC format with protocol files mapping utterance IDs to bonafide/spoof labels and system identifiers.

3.1.2 ASVspoof5 (2024)

ASVspoof5 [8] is the most recent release in the series, collected under more challenging and diverse conditions with a broader range of modern synthesis systems. Including it alongside the 2019 data exposes the model to a wider variety of spoofing artifacts, supporting better generalization.

3.1.3 Fake-or-Real (for-norm subset)

Fake-or-Real [22] is a publicly available dataset comprising real recordings and AI-generated speech. We use only the **for-norm** folder—studio-quality, noise-normalized samples—which

contributes approximately 69,300 samples to Dataset 2 ($\sim 34,650$ real and 34,650 fake), adding diversity in recording conditions and synthesis methods not present in the ASVspoof corpora.

3.2 Dataset Configurations

Table 1: Dataset configurations used in this project.

Configuration	Name	Included datasets			
Dataset 1	ASVspoof Combined (ASVsp-C)	ASVspoof	2019	LA	+
		ASVspoof5	2024		
Dataset 2	ASVspoof-FoR Extended (ASVsp-FoR)	ASVspoof	2019	LA	+
		ASVspoof5	2024	+	Fake-or-Real (for-norm)

3.3 Data Splitting

After merging all entries, the pool is split into train, validation, and test sets using an **80 / 10 / 10** ratio. Class balance is enforced independently for real and fake entries, shuffled with random seed 42 for reproducibility.

Table 2: Sample counts by dataset configuration, split, and class.

Configuration	Split	Real	Fake	Total
Dataset 1	Train	42,139	42,139	84,278
Dataset 1	Dev	5,267	5,267	10,534
Dataset 1	Test	5,268	5,268	10,536
Dataset 2	Train	69,823	69,895	139,718
Dataset 2	Dev	8,727	8,736	17,463
Dataset 2	Test	8,729	8,738	17,467

4 System Architecture and Proposed Methodology

4.1 End-to-End System Overview

FusionGuardNet follows a three-stage pipeline with clear stage separation: (1) offline feature extraction through frozen encoders, (2) learnable feature fusion, and (3) Nes2Net classification. The two encoders are never fine-tuned; only the fusion layer and backbone are trained.

4.2 Audio Signal Preprocessing

4.3 Acoustic and Semantic Feature Extraction: WavLM and Whisper

Both WavLM and Whisper representations are computed once for all clips and saved as PyTorch tensor files (`.pt`), avoiding redundant forward passes through large models during

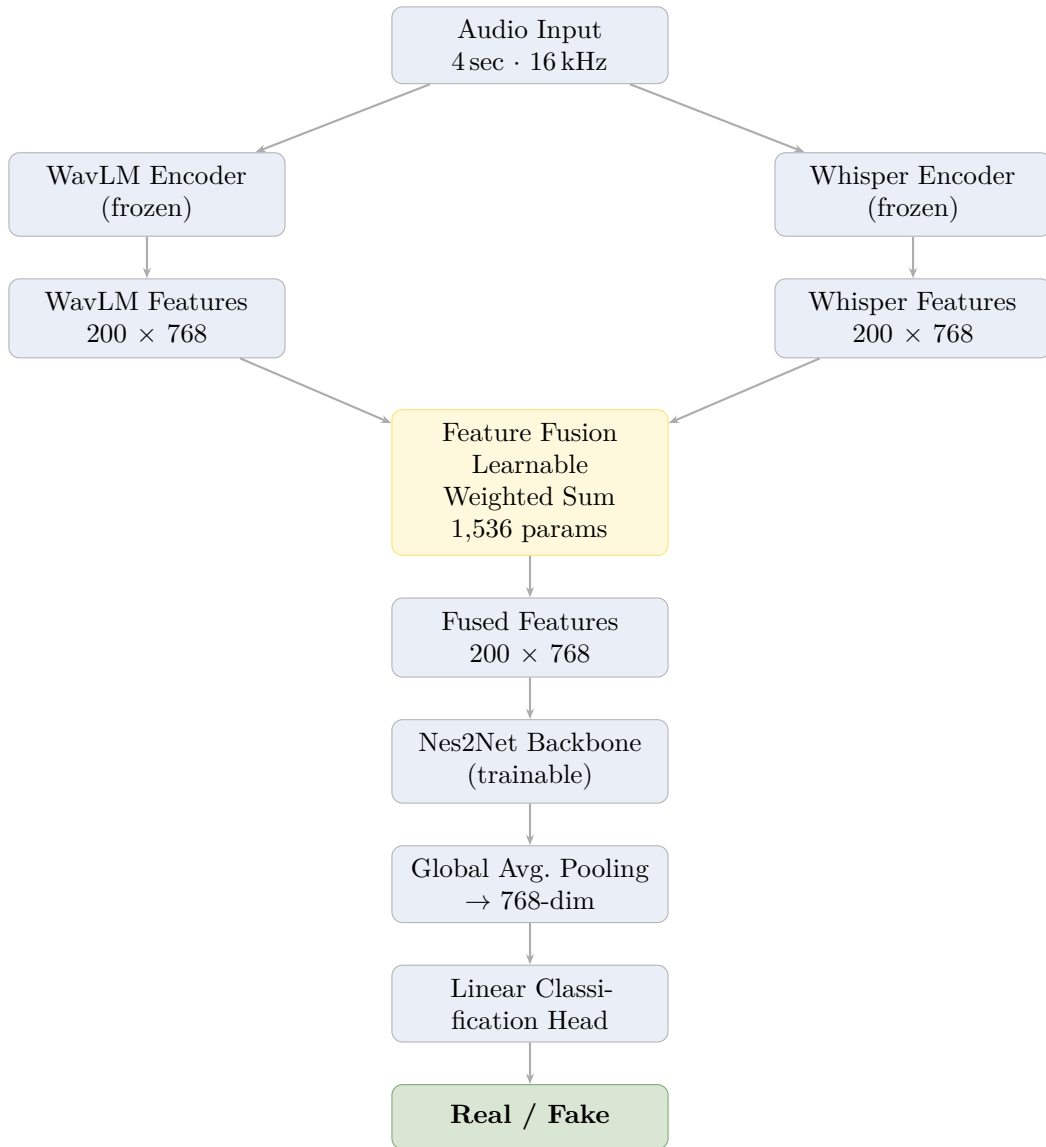


Figure 3: FusionGuardNet end-to-end pipeline. The two frozen encoders extract complementary features; a learnable fusion layer combines them; the Nes2Net backbone classifies the result.

Table 3: Audio preprocessing steps applied uniformly across all source datasets.

Step	Description
Channel reduction	Multi-channel recordings are averaged to mono.
Sample rate normalization	Every waveform is resampled to 16,000 Hz (Torchaudio).
Fixed-length normalization	Normalized to exactly 4 seconds (64,000 samples). Longer clips are randomly cropped; shorter clips are zero-padded at the end.
Padding-validity mask	A frame-level binary mask marks real frames as 1 and padded frames as 0.

training.

4.3.1 WavLM extraction

The preprocessed 16 kHz mono waveform is passed into the frozen `microsoft/wavlm-base-plus` encoder, producing **768-dimensional** frame-level embeddings at 20 ms resolution. A 4-second clip yields **200 frames**.

4.3.2 Whisper extraction

The waveform is first converted to an **80-band log-Mel spectrogram** and passed through the frozen `openai/whisper-small` encoder (decoder discarded). The encoder produces **768-dimensional** contextual embeddings reflecting phonetic and prosodic structure.

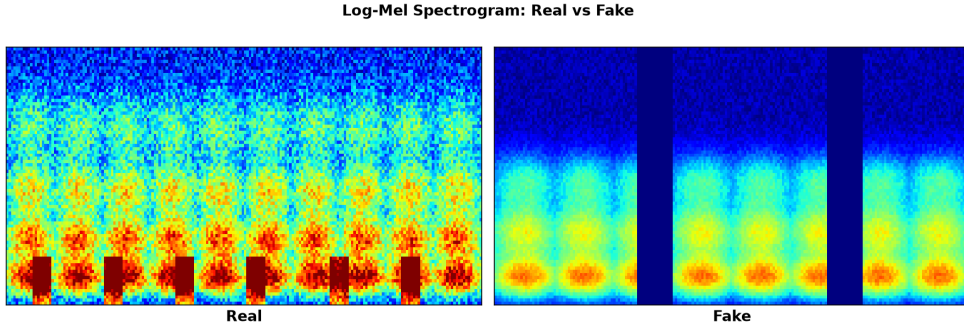


Figure 4: Log-Mel spectrogram comparison. Real speech (left) shows denser, more irregular low-frequency energy; synthetic speech (right) displays smoother, more uniform spectral structure with visible silence bands—differences that motivate the Whisper encoder branch.

4.3.3 Temporal alignment

WavLM and Whisper produce feature sequences of different temporal lengths for the same 4-second input. The Whisper sequence is aligned to the WavLM sequence length by truncation (if longer) or zero-padding at the end (if shorter), ensuring a shared time axis for frame-by-frame fusion.

4.4 Feature Fusion: Combining WavLM and Whisper

The fusion module takes two aligned sequences of shape $T \times 768$ and combines them via a **learnable weighted sum**. Two weight vectors $\alpha_1, \alpha_2 \in \mathbb{R}^{768}$ are learned; before combining, they are passed through a softmax so the weights are positive and sum to 1 at each feature dimension:

$$\mathbf{f}_t = \sigma_1 \odot \mathbf{w}_t + \sigma_2 \odot \mathbf{s}_t, \quad [\sigma_1, \sigma_2] = \text{softmax}([\alpha_1, \alpha_2]) \quad (1)$$

where $\mathbf{w}_t$ and $\mathbf{s}_t$ are the WavLM and Whisper embeddings at time t , and $\odot$ denotes element-wise multiplication. This adds only $2 \times 768 = 1,536$ trainable parameters and is interpretable: the learned weights reveal which model the classifier relies on per feature dimension.

4.5 Classification Model: Adapting Nes2Net to the Fused Features

The fused $T \times 768$ sequence is passed directly to the Nes2Net backbone without dimensionality reduction. After the stack of nested residual blocks, global average pooling collapses the temporal dimension to a single 768-dim vector, which is passed to a linear classification head producing two logits (real/fake). During training, cross-entropy loss is applied; at inference, the argmax determines the final prediction.

The NES ratio is (8, 8) and dilation factor is 2, matching the original Nes2Net configuration [33]. A dropout rate of 0.5 is applied during training.

4.6 Feature Dimensions and Architectural Comparison

Table 4: Tensor dimensions at each processing stage of FusionGuardNet.

Component	Output shape	Explanation
Input audio	64,000 samples	4 sec at 16 kHz
WavLM features	200×768	Frame-level, 20 ms stride
Whisper (pre-alignment)	variable $\times 768$	Encoder embeddings from log-Mel
Whisper (post-alignment)	200×768	Truncated or zero-padded
Fusion input	$2 \times 200 \times 768$	WavLM + Whisper branches
Fusion output	200×768	Channel-wise weighted combination
Fusion trainable params	1,536	2 weight vectors of dim 768
Nes2Net input	200×768	No projection layer needed
Global pooling output	768	Temporal axis collapsed
Classification output	2 logits	Real / Fake

5 Experimental Setup

5.1 Environment and Software Libraries

All experiments were run on a CUDA-enabled GPU with automatic CPU fallback. The project uses PyTorch as the main framework, Torchaudio for audio I/O, and the Hugging Face Transformers library to load WavLM and Whisper. Additional libraries include NumPy, Pandas, scikit-learn, Matplotlib, and Seaborn.

5.2 Training Procedure

Training used the Adam optimizer with learning rate 10^{-4} and weight decay 10^{-4} (L2 regularization), cross-entropy loss, gradient clipping (max norm 1.0), and a ReduceLROnPlateau scheduler that halves the learning rate when validation loss stagnates for 2 consecutive epochs. Training ran for **8 epochs** with batch size **16** and early stopping (patience 4). The best checkpoint was saved by combined validation accuracy and loss. Random seed was fixed to 42 for reproducibility.

5.3 Hyperparameter Configuration

Table 5: Hyperparameter configuration used for all training runs.

Hyperparameter	Value
Input feature dimension	768
NES ratio	(8, 8)
Dilation factor	2
Dropout rate	0.5
Pooling	Global average pooling
Audio sample rate	16,000 Hz
Audio duration	4 seconds
Samples per clip	64,000
Temporal sequence length	200 frames
Fusion type	Learnable weighted sum (softmax-normalised)
WavLM variant	microsoft/wavlm-base-plus
Whisper variant	openai/whisper-small
Optimizer	Adam
Learning rate	10^{-4}
Weight decay	10^{-4}
Batch size	16
Max epochs	8
Early stopping patience	4

5.4 Evaluation Metrics

Table 6: Evaluation metrics used in this project.

Metric	Purpose
Accuracy	Overall fraction of correctly classified samples.
Cross-Entropy loss	Tracks convergence and overfitting.
Precision	Fraction of predicted fakes that are truly fake.
Recall	Fraction of actual fakes correctly detected.
F1-score	Harmonic mean of precision and recall.
EER	Threshold-independent metric: operating point where false acceptance rate equals false rejection rate; lower is better.

6 Results and Discussion

6.1 Baseline Model Performance: WavLM + Nes2Net

The baseline uses WavLM representations fed directly to the Nes2Net classifier, with no Whisper branch. This provides a controlled reference: the classification objective and backbone are identical to the proposed model, so any performance difference is attributable solely to the addition of Whisper-derived information.

6.2 Enhanced Model Performance: WavLM + Whisper + Nes2Net

The enhanced configuration integrates WavLM and Whisper representations before classification, allowing the model to jointly consider low-level acoustic signatures and higher-level phonetic-linguistic patterns.

Table 7: Main quantitative results for FusionGuardNet (WavLM + Whisper + Nes2Net).

Experiment	Accuracy	Precision	Recall	F1-score	EER
Dataset 1	99.18%	99.18%	99.18%	99.18%	N/A
Dataset 2	99.34%	99.12%	99.57%	99.35%	0.60%

For **Dataset 1** (84,278 train / 10,536 test), the model produced only 86 total mistakes with a perfectly symmetric error profile (TN = 5,225, FP = 43, FN = 43, TP = 5,225). Training converged stably over 8 epochs, from 92.60% in epoch 1 to 99.49% in epoch 8, with best dev accuracy of 99.25%.

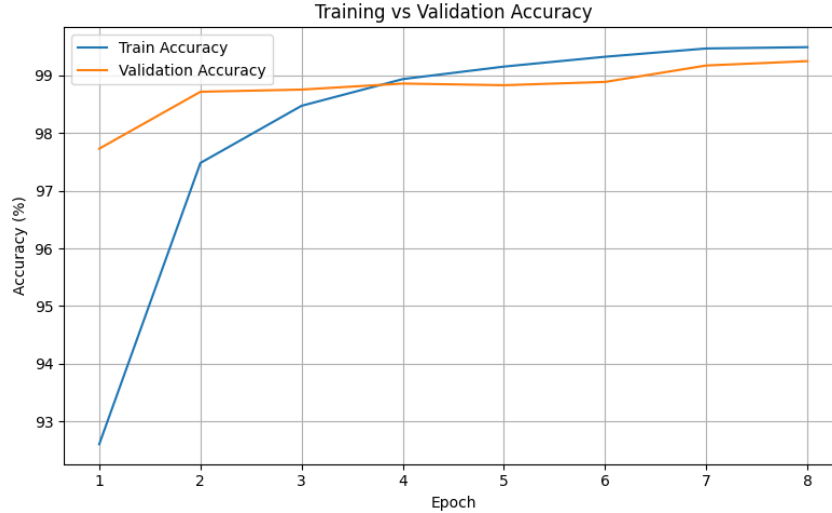


Figure 5: Dataset 1 — Training vs. Validation Accuracy over 8 epochs. Stable convergence to 99.49% training accuracy by epoch 8.

For **Dataset 2** (139,718 train / 17,467 test), the system reached 99.34% accuracy, 0.60% EER, and confusion-matrix counts of TN = 8,652, FP = 77, FN = 38, TP = 8,700.

6.3 Ablation Study: Relative Contribution of Whisper Integration

Table 8: Ablation study: baseline vs. proposed model.

Setting	Feature source	Classifier
Baseline	WavLM only	Nes2Net
Proposed model	WavLM + Whisper	Nes2Net

Because both settings keep the same classifier family and objective, the main changing factor is the addition of Whisper-derived information, making it possible to attribute robustness

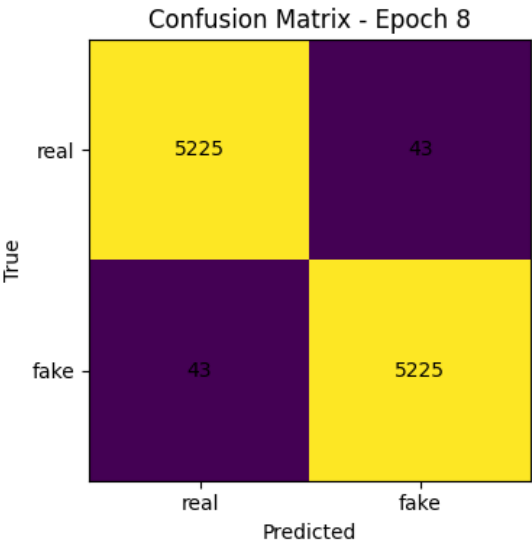


Figure 6: Dataset 1 — Confusion Matrix (Epoch 8). Perfectly symmetric errors: 43 false positives and 43 false negatives out of 10,536 test samples.

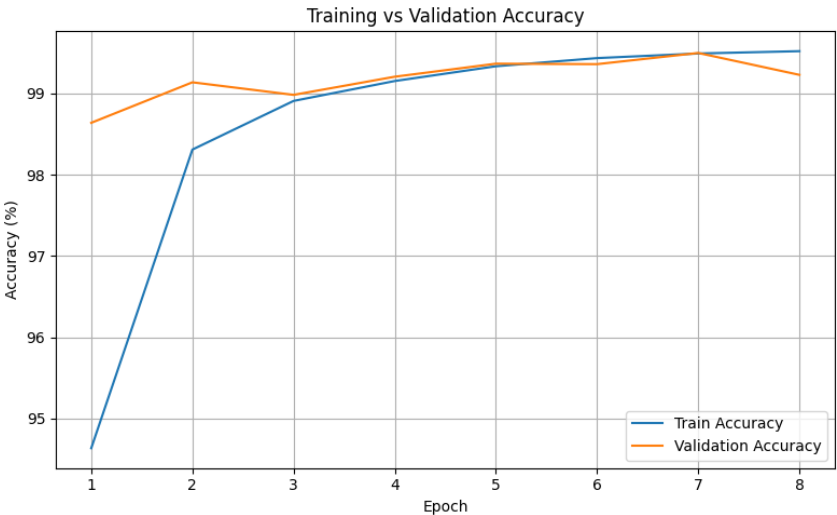


Figure 7: Dataset 2 — Training vs. Validation Accuracy over 8 epochs. The fused model maintains stable convergence on the larger mixed-source dataset.

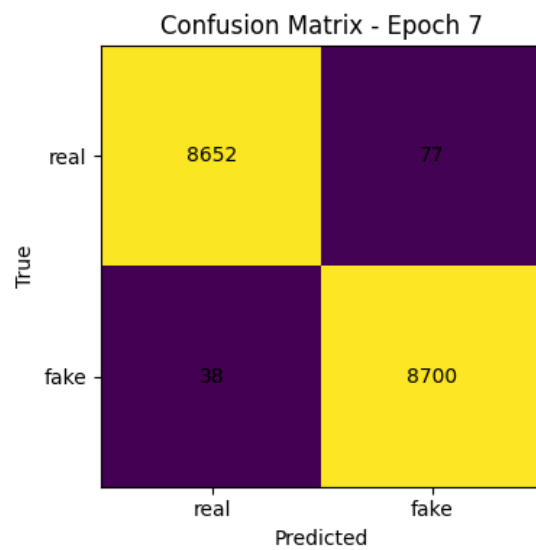


Figure 8: Dataset 2 — Confusion Matrix (Epoch 7). TN = 8,652, FP = 77, FN = 38, TP = 8,700 across 17,467 test samples.

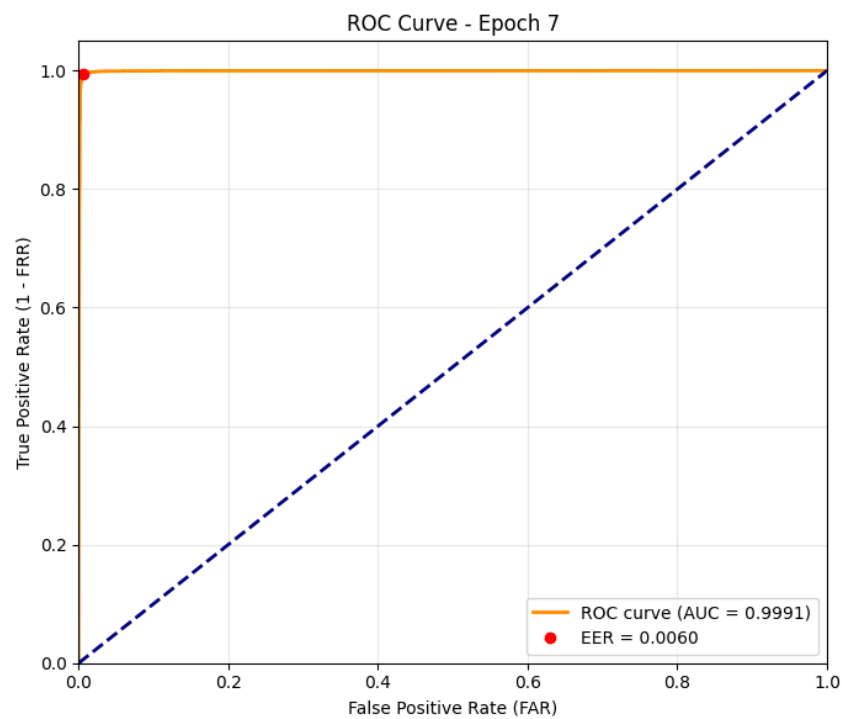


Figure 9: Dataset 2 — ROC Curve (AUC = 0.9991, EER = 0.60%, Epoch 7). Near-perfect discrimination across all operating thresholds.

gains cleanly to multimodal feature integration.

6.4 Error Analysis

In Dataset 1 the error profile is perfectly symmetric ($FP = FN = 43$). In Dataset 2 it is slightly asymmetric ($FP = 77$, $FN = 38$) but strongly balanced relative to dataset size, indicating the model does not strongly favour either class. Most mistakes cluster near the decision boundary or represent genuinely hard synthesis artifacts that confuse both branches. This points toward two practical improvements: (1) calibration-oriented techniques to reduce overconfident mistakes; (2) targeted data expansion for borderline cases.

7 Conclusion and Future Work

7.1 Summary of Achievements

FusionGuardNet implements a complete audio deepfake detection pipeline based on feature-level integration of WavLM and Whisper, followed by a Nes2Net classifier. Both encoders remain frozen; only the 1,536-parameter fusion layer and the backbone are trained. The system achieves over 99% accuracy on both dataset configurations, with a balanced error profile and strong EER, validating the hypothesis that acoustic + phonetic fusion yields a more robust detector than either modality alone.

7.2 Future Research Directions

- **Full baseline suite:** run WavLM-only and Whisper-only settings with identical reporting to directly quantify each branch’s contribution.
- **Robustness evaluation** on additional datasets and unseen synthesis methods to assess generalization under distribution shift.
- **Richer fusion strategies:** attention-based or cross-modal attention fusion instead of the current weighted sum.
- **Calibration methods** to reduce overconfident mistakes on borderline samples.
- **Lightweight inference** for near-real-time screening in production pipelines.
- **Interpretability analysis** of time-frequency regions to support explanation and diagnosis of failure modes.

References

- [1] Li, M., Ahmadiadli, Y., & Zhang, X.-P. (2024). *A Survey on Speech Deepfake Detection*. arXiv:2404.13914. <https://arxiv.org/abs/2404.13914>
- [2] Yi, J., Wang, C., Tao, J., Zhang, X., Zhang, C. Y., & Zhao, Y. (2023). *Audio deepfake detection: A survey*. arXiv:2308.14970. <https://arxiv.org/abs/2308.14970>

- [3] Li, X., Chen, P.-Y., & Wei, W. (2025). *Where are we in audio deepfake detection?* ACM Transactions on Internet Technology, 25(3). <https://arxiv.org/abs/2410.04324>
- [4] Hong, M., Jiang, D., Xie, Z., Zhao, W., Wang, G., & Zhang, C. J. (2026). *Vulnerabilities of audio-based biometric authentication systems against deepfake speech synthesis*. arXiv:2601.02914. <https://arxiv.org/abs/2601.02914>
- [5] Kamel, K., Sood, K., Sankar Dutta, H., & Aryal, S. (2025). *A survey of threats against voice authentication and anti-spoofing systems*. arXiv:2508.16843. <https://arxiv.org/abs/2508.16843>
- [6] Chandra, N. A., et al. (2025). *Deepfake-Eval-2024: A multi-modal in-the-wild benchmark*. arXiv:2503.02857. <https://arxiv.org/abs/2503.02857>
- [7] Kinnunen, T., Yamagishi, J., Todisco, M., Delgado, H., Sahidullah, M., Wang, W., Evans, N., & Lee, K. A. (2019). *ASVspoof 2019: Future horizons in spoofed and fake audio detection*. In *Proc. Interspeech 2019*, 1008–1012. <https://arxiv.org/abs/1904.05441>
- [8] Wang, X., et al. (2024). *ASVspoof 5: Crowdsourced speech data, deepfakes, and adversarial attacks at scale*. In *Proc. ASVspoof 2024 Workshop*. <https://arxiv.org/abs/2408.08739>
- [9] Liu, X., et al. (2023). *ASVspoof 2021: Towards spoofed and deepfake speech detection in the wild*. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 31, 2507–2522. <https://arxiv.org/abs/2210.02437>
- [10] Todisco, M., Delgado, H., & Evans, N. (2017). *Constant Q cepstral coefficients: A spoofing countermeasure for automatic speaker verification*. Computer Speech & Language, 45, 516–535. <https://doi.org/10.1016/j.csl.2017.01.001>
- [11] Muller, N. M., Evans, N., Tak, H., Sperl, P., & Bottinger, K. (2024). *Harder or different? Understanding generalization of audio deepfake detection*. In *Proc. Interspeech 2024*. <https://arxiv.org/abs/2406.03512>
- [12] Nguyen, B., Shi, S., Ofman, R., & Le, T. (2025). *What you read isn't what you hear: Linguistic sensitivity in deepfake speech detection*. arXiv:2505.17513. <https://arxiv.org/abs/2505.17513>
- [13] Zhu, Y., Koppiseti, S., Tran, T., & Bharaj, G. (2024). *SLIM: Style-linguistics mismatch model for generalized audio deepfake detection*. In *Advances in NeurIPS 37*. <https://arxiv.org/abs/2407.18517>
- [14] Shi, H., Shi, X., Dogan, S., Alzubi, S., Huang, T., & Zhang, Y. (2025). *Benchmarking audio deepfake detection robustness in real-world communication scenarios*. arXiv:2504.12423. <https://arxiv.org/abs/2504.12423>
- [15] Chen, S., Wang, C., Chen, Z., Wu, Y., Liu, S., et al. (2022). *WavLM: Large-scale self-supervised pre-training for full stack speech processing*. IEEE Journal of Selected Topics in Signal Processing, 16(6), 1505–1518. <https://arxiv.org/abs/2110.13900>

- [16] Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2023). *Robust speech recognition via large-scale weak supervision*. In *Proc. ICML 2023*, 28492–28518. <https://arxiv.org/abs/2212.04356>
- [17] Yang, S.-W., Chi, P.-H., Chuang, Y.-S., et al. (2021). *SUPERB: Speech processing universal performance benchmark*. In *Proc. Interspeech 2021*, 1194–1198. <https://arxiv.org/abs/2105.01051>
- [18] El Kheir, Y., Samih, Y., Maharjan, S., Polzehl, T., & Moller, S. (2025). *Comprehensive layer-wise analysis of SSL models for audio deepfake detection*. In *Findings of ACL: NAACL 2025*, 4070–4082. <https://arxiv.org/abs/2502.03559>
- [19] Guo, Y., Huang, H., Chen, X., Zhao, H., & Wang, Y. (2024). *Audio deepfake detection with self-supervised WavLM and multi-fusion attentive classifier*. In *Proc. ICASSP 2024*, 12702–12706. <https://arxiv.org/abs/2312.08089>
- [20] Combei, D., Stan, A., Oneata, D., & Cucu, H. (2024). *WavLM model ensemble for audio deepfake detection*. In *Proc. ASVspoof 2024 Workshop*, 170–175. <https://arxiv.org/abs/2408.07414>
- [21] Stourbe, T., Miara, V., Lepage, T., & Dehak, R. (2024). *Exploring WavLM back-ends for speech spoofing and deepfake detection*. In *Proc. ASVspoof 2024 Workshop*, 72–78. <https://arxiv.org/abs/2409.05032>
- [22] Abdel Dayem, M. (n.d.). *The Fake-or-Real Dataset*. Kaggle. <https://www.kaggle.com/datasets/mohammedabdeldayem/the-fake-or-real-dataset>
- [23] Pan, Z., Liu, T., Sailor, H. B., & Wang, Q. (2024). *Attentive merging of hidden embeddings from pre-trained speech model for anti-spoofing detection*. In *Proc. Interspeech 2024*. <https://arxiv.org/abs/2406.10283>
- [24] El Kheir, Y., Das, A., Erdogan, E. E., et al. (2025). *Two views, one truth: Spectral and self-supervised features fusion for robust speech deepfake detection*. arXiv:2507.20417. <https://arxiv.org/abs/2507.20417>
- [25] Ahmadiadli, Y., Zhang, X.-P., & Khan, N. (2025). *Beyond identity: A generalizable approach for deepfake audio detection*. arXiv:2505.06766. <https://arxiv.org/abs/2505.06766>
- [26] Baevski, A., Zhou, H., Mohamed, A., & Auli, M. (2020). *wav2vec 2.0: A framework for self-supervised learning of speech representations*. In *Advances in NeurIPS 33*, 12449–12460. <https://arxiv.org/abs/2006.11477>
- [27] Hsu, W.-N., Bolte, B., Tsai, Y.-H. H., Lakhotia, K., Salakhutdinov, R., & Mohamed, A. (2021). *HuBERT: Self-supervised speech representation learning by masked prediction of hidden units*. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29, 3451–3460. <https://arxiv.org/abs/2106.07447>

- [28] Kawa, P., Plata, M., Czuba, M., Szymanski, P., & Syga, P. (2023). *Improved DeepFake detection using Whisper features*. In *Proc. Interspeech 2023*, 4009–4013. <https://arxiv.org/abs/2306.01428>
- [29] He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep residual learning for image recognition*. In *Proc. CVPR 2016*, 770–778. <https://arxiv.org/abs/1512.03385>
- [30] Gao, S.-H., Cheng, M.-M., Zhao, K., Zhang, X.-Y., Yang, M.-H., & Torr, P. H. S. (2021). *Res2Net: A new multi-scale backbone architecture*. *IEEE Transactions on PAMI*, 43(2), 652–662. <https://arxiv.org/abs/1904.01169>
- [31] Li, X., Li, N., Weng, C., Liu, X., Su, D., Yu, D., & Meng, H. (2021). *Replay and synthetic speech detection with Res2Net architecture*. In *Proc. ICASSP 2021*, 6354–6358. <https://arxiv.org/abs/2010.15006>
- [32] Fan, C., Xue, J., Tao, J., Yi, J., Wang, C., Zheng, C., & Lv, Z. (2023). *Spatial reconstructed local attention Res2Net with F0 subband for fake speech detection*. arXiv:2308.09944. <https://arxiv.org/abs/2308.09944>
- [33] Liu, T., Truong, D. T., Das, R. K., Lee, K. A., & Li, H. (2025). *Nes2Net: A lightweight nested architecture for foundation model driven speech anti-spoofing*. arXiv:2504.05657. <https://arxiv.org/abs/2504.05657>
- [34] Huang, W., Liu, X., Wang, X., Yamagishi, J., & Qian, Y. (2025). *From sharpness to better generalization for speech deepfake detection*. In *Proc. Interspeech 2025*. <https://arxiv.org/abs/2506.11532>
- [35] Kang, T., Han, S., Choi, S., Seo, J., Chung, S., Lee, S., Oh, S., & Kwak, I.-Y. (2024). *Experimental study: Enhancing voice spoofing detection models with wav2vec 2.0*. arXiv:2402.17127. <https://arxiv.org/abs/2402.17127>
- [36] Zhang, Y., Lu, J., Shang, Z., Wang, W., & Zhang, P. (2024). *Improving short utterance anti-spoofing with AASIST2*. In *Proc. ICASSP 2024*. <https://arxiv.org/abs/2309.08279>
- [37] Wu, H., Tseng, Y., & Lee, H.-y. (2024). *CodecFake: Enhancing anti-spoofing models against deepfake audios from codec-based speech synthesis systems*. In *Proc. Interspeech 2024*, 1770–1774. <https://arxiv.org/abs/2406.07237>
- [38] Pascu, O., Stan, A., Oneata, D., Oneata, E., & Cucu, H. (2024). *Towards generalisable and calibrated audio deepfake detection with self-supervised representations*. In *Proc. Interspeech 2024*, 4828–4832. https://www.isca-archive.org/interspeech_2024/pascu24_interspeech.html
- [39] Lopez, J. A., Stemmer, G., & Cordourier Maruri, H. (2025). *Generalizable detection of audio deepfakes*. arXiv:2507.01750. <https://arxiv.org/abs/2507.01750>

A Glossary

Anti-spoofing	Automatically detecting whether a given audio sample is genuine speech or a synthetically generated/modified recording designed to deceive an authentication system.
ASVspoof	A challenge series and dataset collection providing standardized benchmarks for automatic speaker verification anti-spoofing research, running since 2015.
Bonafide	Term used in ASVspoof datasets to label authentic, real human speech samples (as opposed to spoofed samples).
Cross-Entropy Loss	Standard loss function for classification tasks measuring the difference between a predicted probability distribution and the true label distribution.
EER	Equal Error Rate—a threshold-independent metric for binary classifiers measuring the operating point at which false acceptance and false rejection rates are equal; lower is better.
Embedding	A dense, high-dimensional numerical vector produced by encoding an audio frame or utterance through a neural model.
Feature Fusion	Combining feature representations from multiple sources (here, WavLM and Whisper) into a single unified representation before classification.
FLAC	Free Lossless Audio Codec; lossless audio compression format used by ASVspoof 2019 and ASVspoof5 datasets.
HuBERT	Hidden-Unit BERT; a self-supervised speech model from Meta AI that learns representations by predicting discrete cluster assignments for masked audio frames.
Log-Mel Spectrogram	A 2-D time-frequency representation of audio where the frequency axis is Mel-scaled and pixel intensity represents log-energy; the standard input format for Whisper.
Nes2Net	Nested Res2Net; a lightweight backbone for speech anti-spoofing that nests an inner residual shortcut inside each Res2Net block.
Offline Feature Extraction	A pipeline strategy where large model inference is performed once before training and results are saved to disk.
Precision	Fraction of predicted-positive (fake) samples that are truly positive. High precision means few false alarms.

Pre-trained Model

A neural model trained on a large external dataset whose learned weights are reused for a downstream task.

Recall

Fraction of actual positive (fake) samples correctly identified. High recall means few missed detections.

Res2Net

A backbone architecture capturing features at multiple granularities within a single residual block via hierarchical channel splits.